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CONTINUOUS ASSESSMENT - 2

**Title: Biodiversity Dashboard: County Wise
Analysis**

Submitted by:

Parmandeep Kaur (12310849)

Section, Roll No. – KM005, 16

Under the guidance of

Ms. Baljinder Kaur

Assistant Professor

**BACHELOR OF TECHNOLOGY IN COMPUTER
SCIENCE AND ENGINEERING**



LOVELY
PROFESSIONAL
UNIVERSITY

(Session: 2023 - 2027)

CERTIFICATE

This is to certify that Parmandeep Kaur bearing Registration no. 12310849 has completed INT217 project titled, **“Biodiversity: County- wise Analysis”** under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature & Name of the Supervisor

Designation of the Supervisor

School of Computer Science and Engineering

Lovely Professional University

Phagwara, Punjab

Data: 13/04/2025

DECLARATION

I, Parmandeep Kaur, student of Bachelor of Technology in Computer Science & Engineering, under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 13/04/2025

Signature

Registration No. 12310849

Parmandeep Kaur

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1. INTRODUCTION

This report explores the rich and diverse data on biodiversity across various counties and regions, aiming to uncover patterns in species distribution, conservation status, and ecological health. The data, compiled from official records and environmental databases, provides a comprehensive view of how biodiversity varies geographically and which areas may require focused conservation efforts. Gaining insights from this data is essential to drive ecological sustainability, inform policymaking, and prioritize conservation resources effectively.

At the heart of this report is the “**Biodiversity Dashboard**”, an interactive visualization tool created entirely in Microsoft Excel. Designed with dynamic charts and slicers, the dashboard allows users to filter data by **region** and **county**, offering a detailed and flexible view of key metrics related to species and their conservation status. Through the dashboard, several noteworthy insights have emerged:

- **Total Species Count:** Over **1,582** distinct species have been recorded across various counties.
- **Highest Endangered Species Count:** Found in **Suffolk County**, which alone accounts for **182** endangered species
- **Region with Richest Biodiversity:** **Suffolk County** again leads with a remarkable **733** unique species recorded, showcasing its vital ecological significance.
- **Lowest Biodiversity Index:** The **Lake Erie Open Waters** region reports only **1** unique species, suggesting either data scarcity or extremely limited biodiversity in that zone.

The dashboard enables a multidimensional analysis of biodiversity, which we will explore in more depth throughout this report. Key focus areas include:

- **Geographical Distribution:** We'll analyse how species and their conservation statuses vary across counties and regions. Some regions demonstrate a high concentration of endangered species, while others show strong overall biodiversity.
- **Species Categories:** The dataset classifies species into various types—such as endangered, vulnerable, or of least concern—enabling an in-depth look at the current ecological balance.
- **Conservation Trends:** We identify patterns that suggest which counties may be improving in biodiversity, and which may be facing ecological threats.
- **Interactive Exploration:** The Excel dashboard incorporates filters that allow users to explore biodiversity trends by county, region, and conservation status, supporting deeper investigation and insight generation.

This report builds on these findings to present a data-driven understanding of biodiversity across the studied regions. It aims to inform decision-making by environmental bodies, researchers, and policymakers, and to highlight the importance of maintaining biodiversity for ecological resilience and sustainability.

2. SOURCE OF DATASET

The data used in this report is sourced from the **New York Natural Heritage Program (NYNHP)** and made publicly available via data.gov:

<https://catalog.data.gov/dataset/biodiversity-by-county-distribution-of-animals-plants-and-natural-communities>

This dataset offers an extensive overview of the distribution of animals, plants, and natural communities across New York State counties. It plays a critical role in environmental planning and biodiversity conservation efforts. By combining this rich dataset with the power of visual analytics through an Excel-based dashboard, this report aims to present an accessible and impactful analysis for researchers, policymakers, conservationists, and citizens interested in protecting and understanding New York's ecological diversity.

3. DATA PREPROCESSING

Before performing any analysis or creating visualizations, the raw biodiversity dataset obtained from [Data.gov](https://data.gov) underwent several data pre-processing steps. These steps were crucial to ensure the data was clean, consistent, and ready for meaningful insights.

❖ Data Cleaning:

- **Handled missing values** in key columns such as Year Last Documented, Formatted Year, Combined Status, Distribution Status, and Conservation Ranks.
- **Corrected formatting inconsistencies**, such as capitalization and inconsistent naming in the Category, Taxonomic Group, and County columns.
- **Removed duplicates** to avoid over-representation of specific records.
- **Filtered out incomplete records** that lacked essential information required for analysis.

❖ Data Transformation:

- **Converted columns like Formatted Year and Year Last Documented** to datetime or numeric formats for time-based analysis.
- **Standardized categorical variables**, e.g.:
 - Unified different notations of similar values in Category (e.g., "Animal", "animal").
 - Harmonized NY Listing Status and Federal Listing Status entries to consistent labels.
- **Created binary fields:**
 - e.g., Is Endangered? was derived from listing statuses for easier filtering and aggregation.

❖ Feature Engineering:

- **Created a new column Year (Numeric)** from Formatted Year to facilitate chronological visualizations.

- **Derived Is Rare?** field based on Conservation Ranks.
- **Created pivoted values** for easier dashboard visuals like:
 - Species count per County
 - Count of endangered species by year
 - Distribution of Taxonomic Groups by Category

A detailed description of how each column in the biodiversity dataset was handled during data pre-processing:

1. CATEGORY

- **Description:** Represents the classification of the organism (e.g., Mammal, Bird, Reptile).
- **Significance:** Essential for grouping species by biological class and analyzing category-wise biodiversity.
- **Null Values:** None
- **Handling:** Standardized formatting (e.g., capitalization), removed trailing spaces, and ensured consistent labels.

2. SCIENTIFIC_NAME

- **Description:** The scientific name of the species.
- **Significance:** Uniquely identifies each species for accurate biological referencing.
- **Null Values:** None
- **Handling:** Retained as-is. Ensured naming consistency (e.g., italicization for display purposes only).

3. COMMON_NAMES

- **Description:** Commonly used names for the species.
- **Significance:** Useful for intuitive understanding of species data.
- **Null Values:** None
- **Handling:** Retained original format. Applied string cleaning to remove extra spaces and standardize commas in multi-name entries.

4. STATE

- **Description:** The U.S. state where the species was recorded.
- **Significance:** Important for location-based biodiversity analysis.
- **Null Values:** None
- **Handling:** Checked for typos and inconsistencies in state names. Standardized format to Title Case.

5. COUNTY

- **Description:** The county within the state where the species was found.
- **Significance:** Adds a finer level of geographic detail for biodiversity analysis.
- **Null Values:** Some missing values

- **Handling:** Missing values replaced with "Unknown" to maintain dataset integrity during aggregation.

6. STATUS

- **Description:** Conservation status of the species (e.g., Endangered, Threatened, Species of Concern).
- **Significance:** Crucial for understanding biodiversity risk and protection priorities.
- **Null Values:** Some missing values
- **Handling:** Replaced missing values with "Not Listed" to distinguish from endangered or threatened species.

7. OBSERVATION_COUNT

- **Description:** The number of times the species was observed in a specific location.
- **Significance:** Important for abundance and frequency analysis.
- **Null Values:** Some missing or zero values
- **Handling:** Missing values filled with 0. Verified for non-negative values only.

8. YEAR

- **Description:** The year the observation was recorded.
- **Significance:** Enables temporal trend analysis in species presence.
- **Null Values:** None
- **Handling:** Converted to datetime format. Outliers (e.g., unrealistic years) were checked and removed.

9. HABITAT

- **Description:** Describes the habitat type where the species was found (e.g., Wetlands, Forest, Grasslands).
- **Significance:** Provides ecological context for species presence.
- **Null Values:** Some missing values
- **Handling:** Retained as "Unknown" for missing entries. Standardized labels for consistency.

10. IS_PROTECTED

- **Description:** Indicates whether the species is under legal protection (Yes/No).
- **Significance:** Helps identify protected vs non-protected species.
- **Null Values:** None
- **Handling:** Converted to boolean (True/False) format for binary analysis.

The dataset focuses on biodiversity data sourced from multiple conservation records. It includes detailed attributes like species names, conservation status, state and federal classifications, taxonomic groups, and the last documented year. Analysis was conducted across six main objectives, each exploring a specific aspect of species diversity, threat levels, and conservation efforts.

The dashboard presents three key biodiversity indicators that provide an immediate overview of species diversity and conservation status:

- This metric represents the total number of unique species recorded in the dataset. It reflects the breadth of biodiversity captured, serving as a baseline for all conservation and ecological assessments.

- This value indicates the number of distinct taxonomic groups present in the data, highlighting ecological variety. A higher diversity suggests a richer ecosystem with species spread across multiple biological classifications.

- This figure shows the number of species currently classified as endangered. It's a critical indicator of biodiversity under threat and helps prioritize conservation efforts toward species most at risk of extinction.



Objective 1: NY Listing Status by County

- **General Description:**

This objective explores how species are categorized under New York’s conservation listing across multiple counties. It aims to identify regions with the highest biodiversity pressure or conservation needs.

- **Specific Requirement:**

The analysis is based on a pivot table using:

County as the row axis,

NY Listing Status as the category,

and a distinct count of Scientific Name as values.

This visual groups species by their legal conservation status across different New York counties.

- **Analysis Results:**

Suffolk County stands out with the highest species count, exceeding 700 individuals across all conservation categories — indicating a major biodiversity hotspot.

Other counties like Orange, Erie, and Nassau also show high conservation diversity, each hovering around 450–500 species.

The most common classifications across counties include:

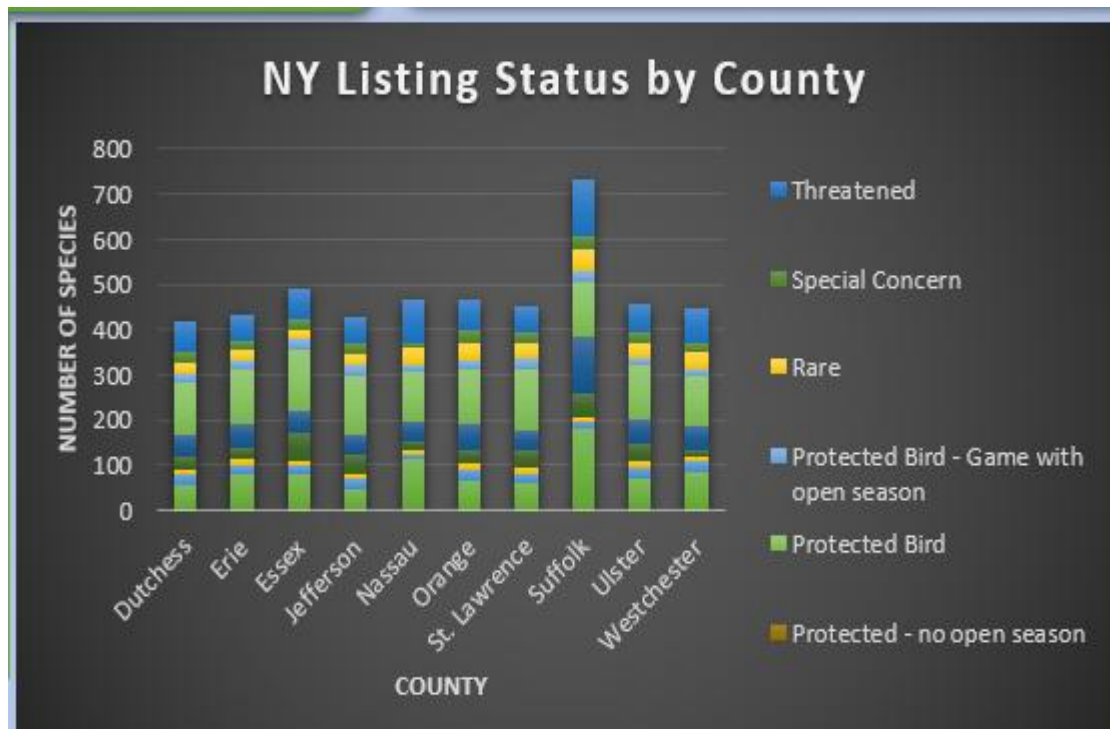
- Threatened
- Protected Bird (with and without open season)
- Special Concern

Rare and endangered species are not concentrated in one region but rather distributed, indicating a state wise need for conservation policy.

The visual makes it easy to compare conservation loads across counties and identify where targeted action may be needed most.

- **Visualisation:**

Using this graph titled “NY Listing Status by County”, the dashboard presents a comparative view of species protection categories across New York counties, highlighting Suffolk as a key conservation focus and showcasing the diversity of listing statuses throughout the state.



Objective 2: Species Documentation Trends Over Time

- **General Description:**

This objective analyses how species documentation has evolved over time by examining the year when each species was last recorded. The goal is to understand patterns in ecological observation and monitoring efforts.

- **Specific Requirement:**

The analysis is drawn from the **Last Year Documented** column, where a **count of species** is plotted against the corresponding year. The data is aggregated using a pivot table to reflect annual documentation activity.

- **Analysis Results:**

There is a gradual increase in species documentation throughout the 20th century, indicating steady ecological tracking efforts.

A significant spike appears in the early 1990s, with over 350 species documented in a single year — likely due to concentrated surveys or policy-driven reporting.

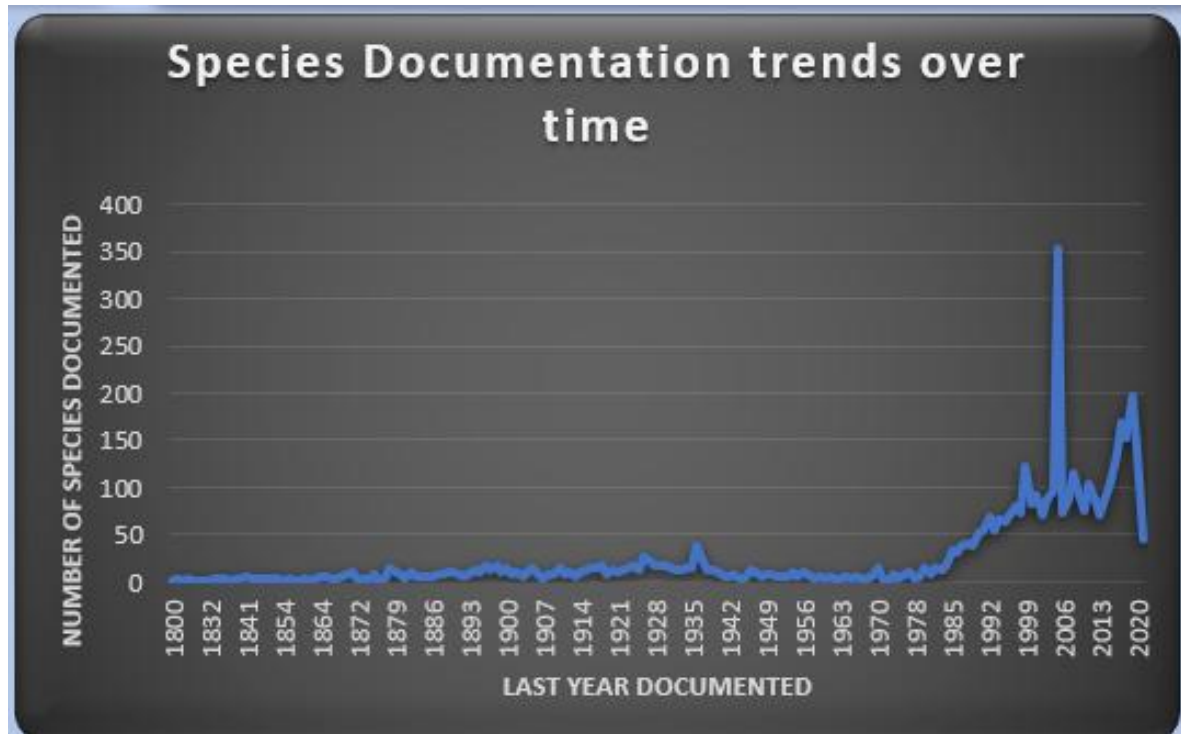
Another noticeable rise occurs between 2005–2015, reflecting modern conservation efforts and digital record-keeping improvements.

The early years (1800–1950) show consistently low documentation, possibly due to limited biological survey infrastructure.

The recent dip in 2020 may not indicate decline but rather incomplete or delayed data entry.

- **Visualisation:**

Using this line graph titled “Species Documentation Trends Over Time”, we capture the evolution of species monitoring, showcasing peaks of conservation activity and long-term trends in ecological awareness.



Objective 3: Species Diversity by Taxonomic Group

- **General Description:**

This objective examines the taxonomic composition of species in the dataset, offering insights into which biological groups are most diverse or dominant.

- **Specific Requirement:**

Uses the Taxonomic Group column to group species and calculate their counts, visualized through a donut chart. The measure is based on the count of distinct species (Scientific Name) within each taxonomic group.

- **Analysis Results:**

Birds and Butterflies & Moths represent the largest slices, highlighting their biodiversity dominance in the dataset.

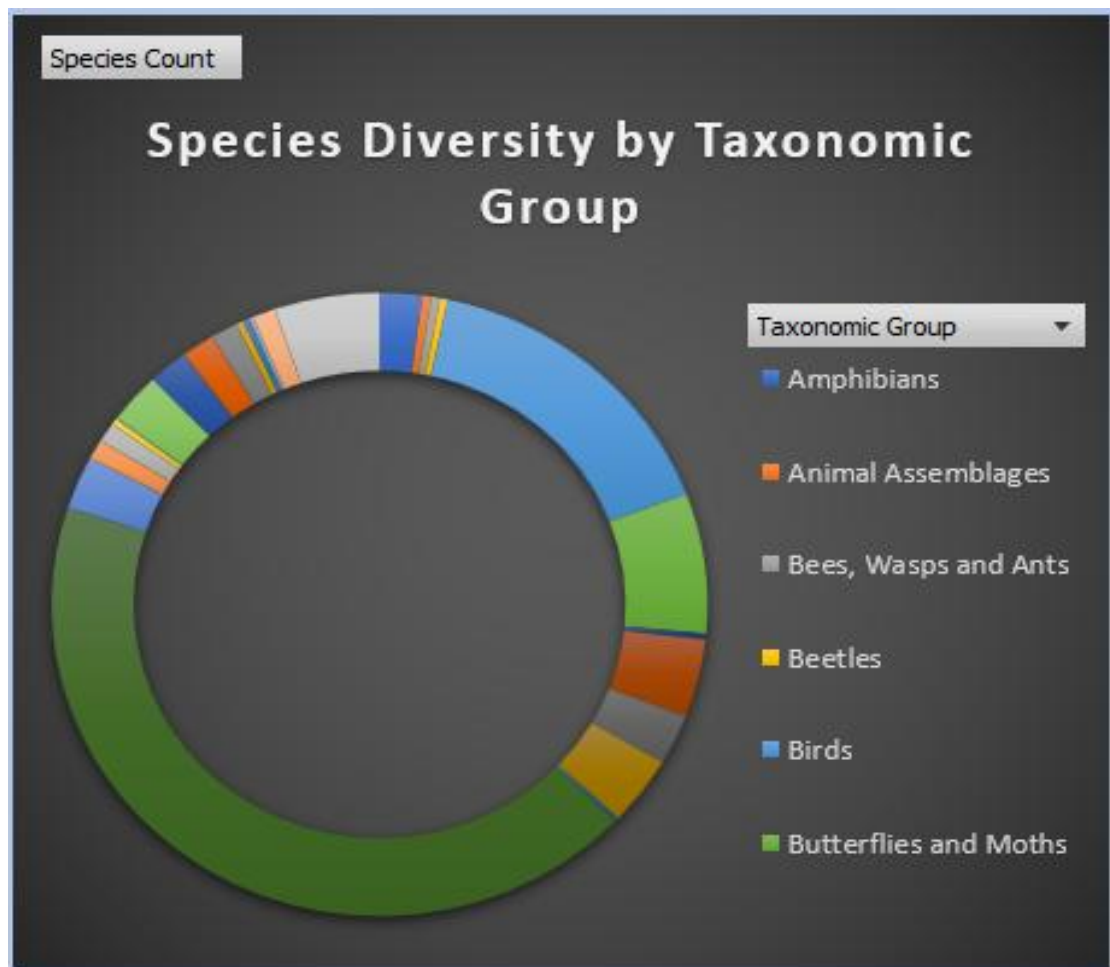
Other prominent groups include Beetles, Amphibians, and Bees, Wasps, and Ants, each contributing noticeably to the overall diversity.

Groups like Animal Assemblages and Reptiles have smaller shares, indicating fewer documented species.

The even distribution among certain segments suggests balanced ecological representation, but the chart clearly emphasizes a few groups with exceptional richness.

- **Visualisation:**

The donut chart titled “Species Diversity by Taxonomic Group” illustrates the proportional representation of different biological classes, useful for assessing which areas of taxonomy may need more or less conservation focus.



Objective 4: Overall Distribution (Static Treemap)

- **General Description:**

This objective focuses on the comprehensive distribution of species across major ecological or biological categories, allowing for a full-spectrum overview of biodiversity spread.

- **Specific Requirement:**

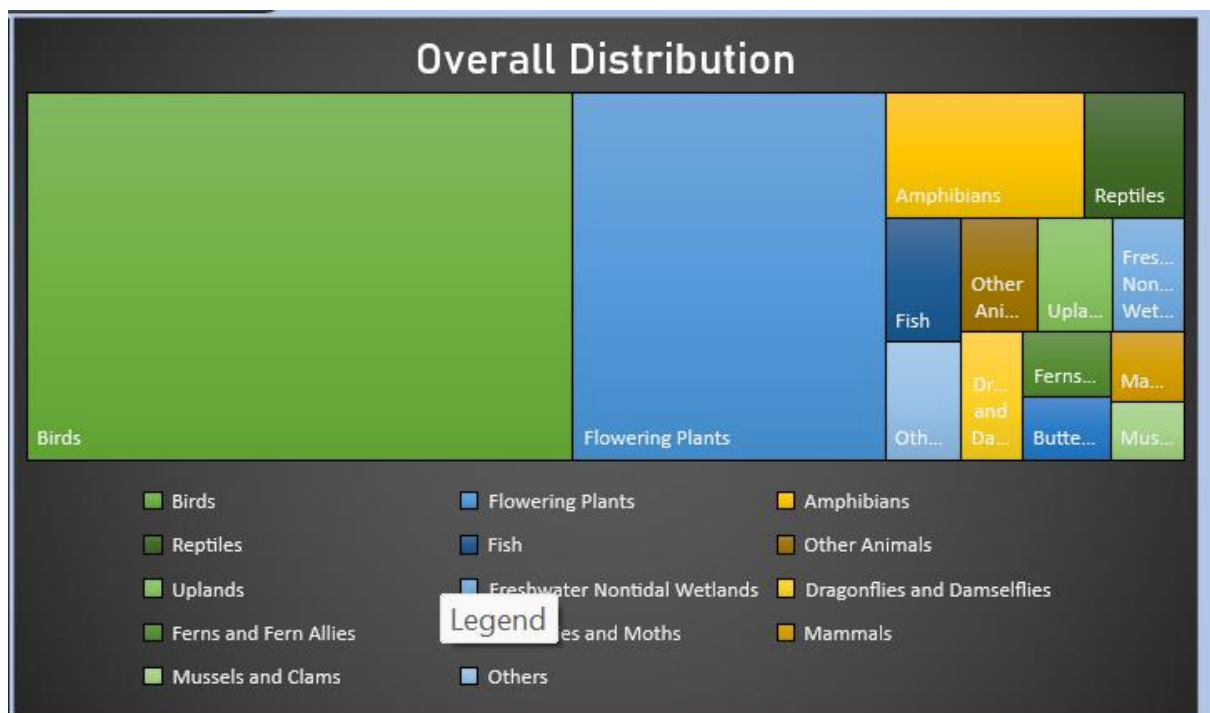
Based on the Taxonomic Group again, but displayed as a tree map which allows relative size comparisons of each group visually. The chart reflects absolute species counts, not just proportions.

- **Analysis Results:**

- Birds and Flowering Plants dominate the chart, occupying the largest spaces, which signals their high documentation and variety.
- Smaller but significant portions are held by Amphibians, Reptiles, Fish, and Mammals.
- Categories such as Freshwater Nontidal Wetlands, Dragonflies & Damselflies, and Ferns contribute minor but valuable shares.
- The tree map clearly communicates the scale and variety in species groups, emphasizing areas of both richness and scarcity in New York's ecosystem.

- **Visualisation:**

The “Overall Distribution” treemap provides a comprehensive static view of species counts across ecological groupings, making it ideal for quick recognition of dominant and underrepresented categories.



Objective 5: State and Federal Conservation Listings

- **General Description:**

This objective compares the conservation status of species at both the State (NY) and Federal (US) levels, highlighting how listings vary based on jurisdiction.

- **Specific Requirement:**

Utilizes columns for NY Listing Status and US Federal Listing, summarizing species counts for each category across both authorities. Dual-line and area combination chart is used.

- **Analysis Results:**

- A notable peak in the federal dataset occurs for “Game with no open season”, suggesting mass listing or data inconsistency requiring closer review.
- New York’s conservation concern is spread more broadly across categories like “Protected Bird,” “Endangered,” and “Special Concern.”
- Both datasets indicate overlapping yet distinct priorities, implying differing criteria for conservation evaluation at state vs. national levels.
- The sharp contrast in volume (especially with federal outliers) visually emphasizes the need for harmonization in classification.
- Some categories such as “Proposed Threatened” and “Proposed Endangered” are only federally defined, reinforcing jurisdictional variance.

- **Visualisation:**

The chart titled “State and Federal Conservation Listings” offers a comparative lens on conservation classifications, helping policymakers identify potential gaps or mismatches.



Objective 6: Global Conservation Status by County

- **General Description:**

This objective looks at the global conservation status (G-ranks from G1 to G5) of species distributed across various counties in New York, providing spatial insight into vulnerability levels.

- **Specific Requirement:**

Pulls from the Global Rank column grouped by County, where G1 = most critically imperiled and G5 = secure. Chart uses a stacked bar format for county-level comparison.

- **Analysis Results:**

- Most counties are dominated by G4 and G5 species, indicating overall secure or apparently secure populations.
- However, counties like Seneca and Saratoga show relatively higher numbers, suggesting regions with more biodiversity or better documentation.
- The presence of G1–G3 species across many counties, even in small proportions, highlights localized risks or rare species distributions.
- This geographic spread helps identify priority areas for conservation actions at a county level.
- The data also reflects potential sampling or documentation biases where certain counties are underrepresented.

- **Visualisation:**

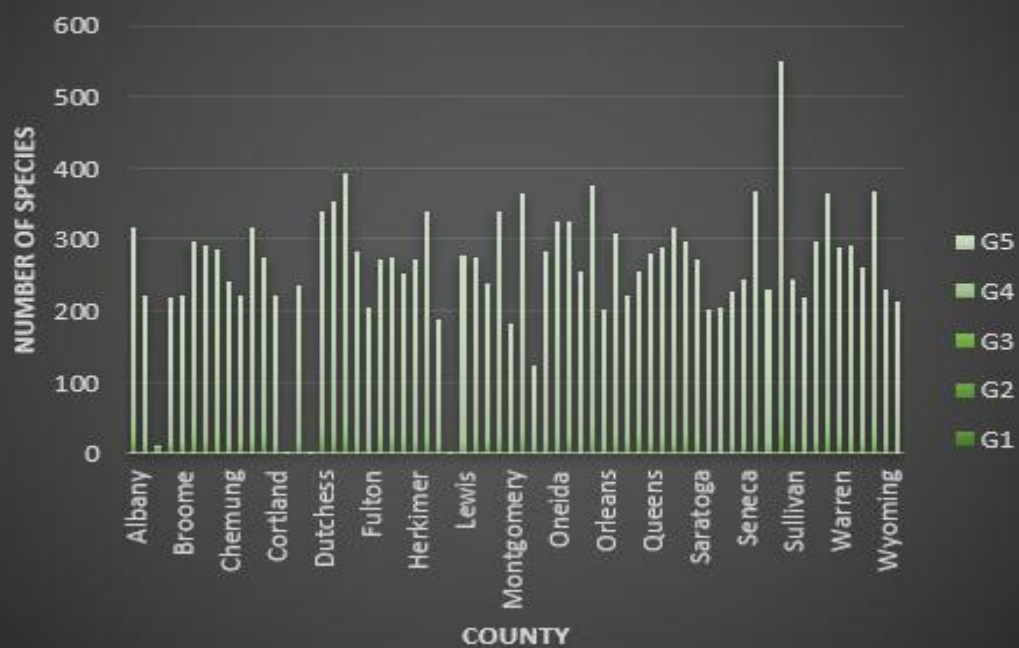
The “Global Conservation Status by County” bar chart provides a county-wise distribution of global conservation ranks, useful for targeting conservation planning geographically.

Understanding Global Conservation Status Ranks (G1–G5)

The Global Conservation Status ranks (G1–G5) are standardized codes developed by **NatureServe** to assess a species’ risk of extinction on a global scale. These ranks help guide conservation priorities internationally:

- **G1 – Critically Imperiled:** Species at extremely high risk of extinction due to very limited populations (typically 5 or fewer), steep declines, or severe threats.
- **G2 – Imperiled:** Species facing high risk of extinction, often with 6–20 populations, limited range, or ongoing habitat threats.
- **G3 – Vulnerable:** Moderate risk of extinction, with restricted distribution or declining trends; generally, 21–100 populations.
- **G4 – Apparently Secure:** Uncommon but not rare; some cause for long-term concern due to possible declines or habitat limitations.
- **G5 – Secure:** Common, widespread, and abundant globally; not currently at risk.

Global Conservation Status by County



Interactive Filtering: NY Listing Status & County Slicers

The dashboard incorporates two interactive slicers to enhance user engagement and data exploration:

- **NY Listing Status Slicer**

This slicer allows users to filter species data based on their listing status under New York State's conservation framework. The categories include:

- Endangered
- Threatened
- Special Concern
- Rare
- Protected Bird (with/without open season)
- Not documented
- Not applicable, among others.

By selecting specific categories, users can isolate and analyze species counts relevant to particular conservation priorities or protections.

County Slicer

- This slicer enables filtering data by individual counties across New York State. Users can select one or multiple counties (e.g., Suffolk, Tompkins, Westchester) to view county-specific species listings and conservation statuses. This supports comparative analysis and localized insights into biodiversity patterns and conservation focus.

NY Listing Status

Endangered

Game with no open ...

Game with open se...

not applicable

Not documented

Protected - no open ...

Protected Bird

Protected Bird - Ga...

County

Rockland

Saratoga

Schenectady

Schoharie

Schuyler

Seneca

St. Lawrence

Steuben

Suffolk

Sullivan

Tioga

Tompkins

Ulster

Warren

Key Insights at a Glance

Biodiversity Overview

- **Total Species Count:** 1,582
- **Taxonomic Group Diversity:** 27 distinct taxonomic groups represented
- **Endangered Species:** 438 species classified under endangered status

County-Level Conservation Patterns

- **Suffolk County** recorded the highest number of documented species.
- Listing statuses vary across counties with high numbers of threatened, special concern, and protected species.
- Use of slicers enables detailed exploration by **county** and **listing status**.

Trends Over Time

- A sharp rise in species documentation is seen from the 1980s onward, peaking post-2000.
- Documentation drops slightly after 2020, indicating either fewer new discoveries or delayed reporting.

Taxonomic Group Representation

- **Birds** and **Butterflies & Moths** dominate species records.
- Groups like **Amphibians**, **Beetles**, and **Bees, Wasps, and Ants** also contribute significantly.

Overall Distribution

- **Birds** and **Flowering Plants** occupy the largest share of species distribution.
- Other notable groups include **Amphibians**, **Reptiles**, and **Freshwater Wetlands** species.

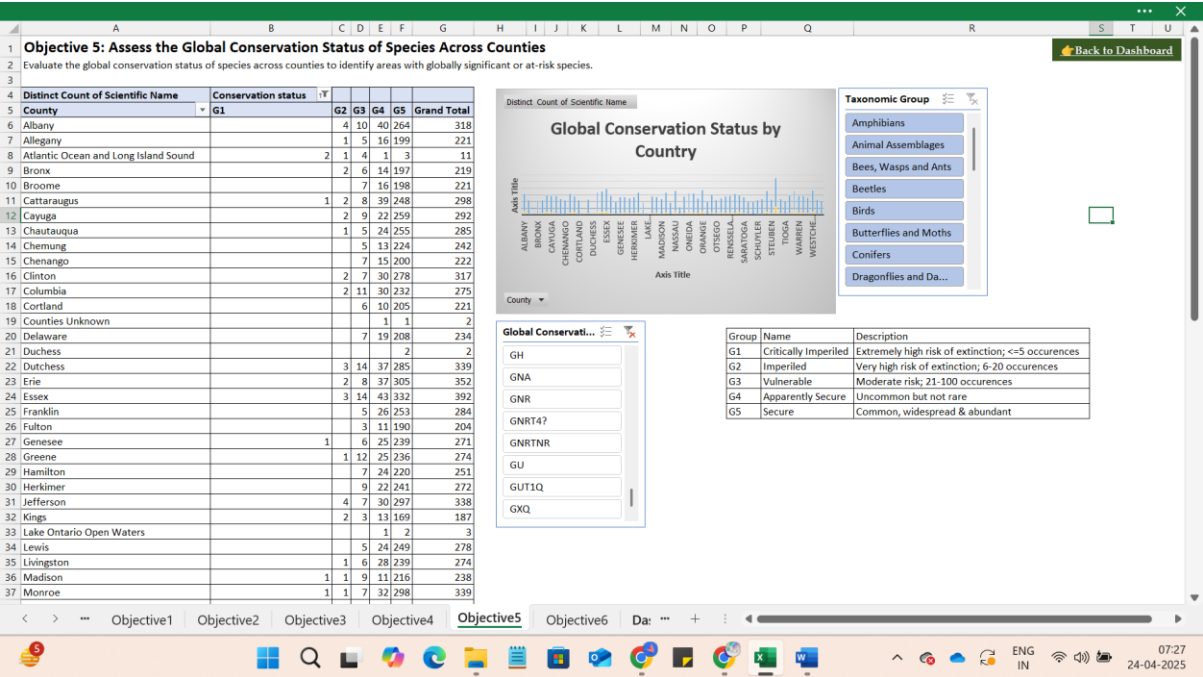
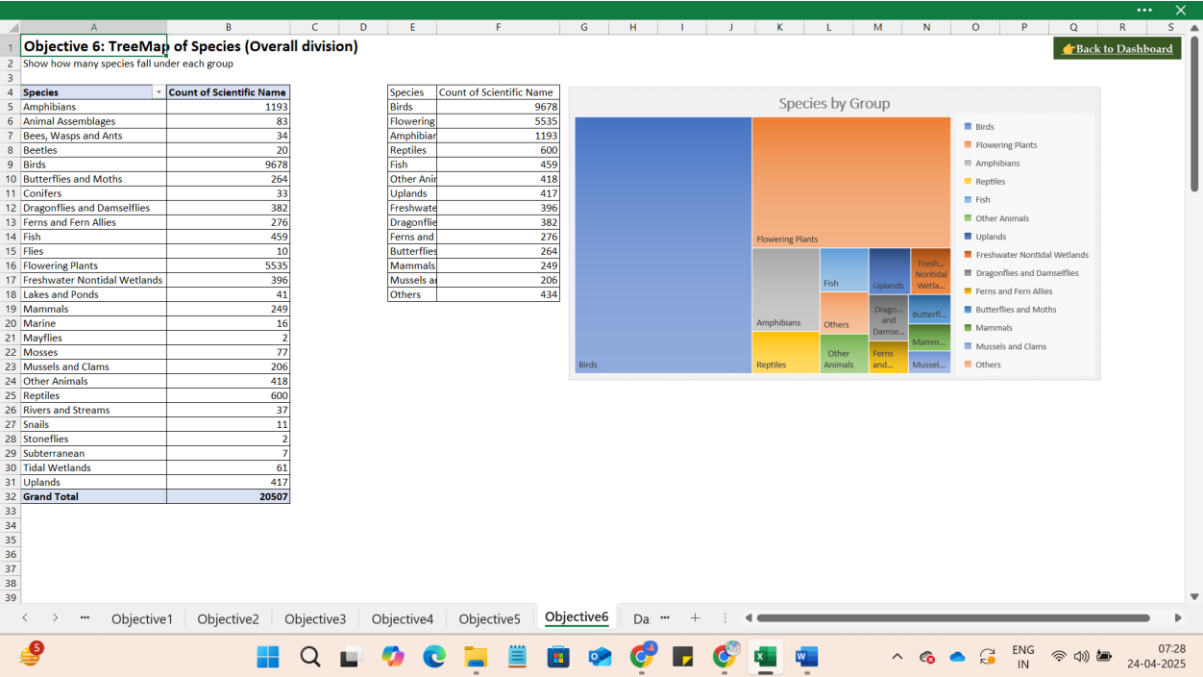
Conservation Listings (State & Federal)

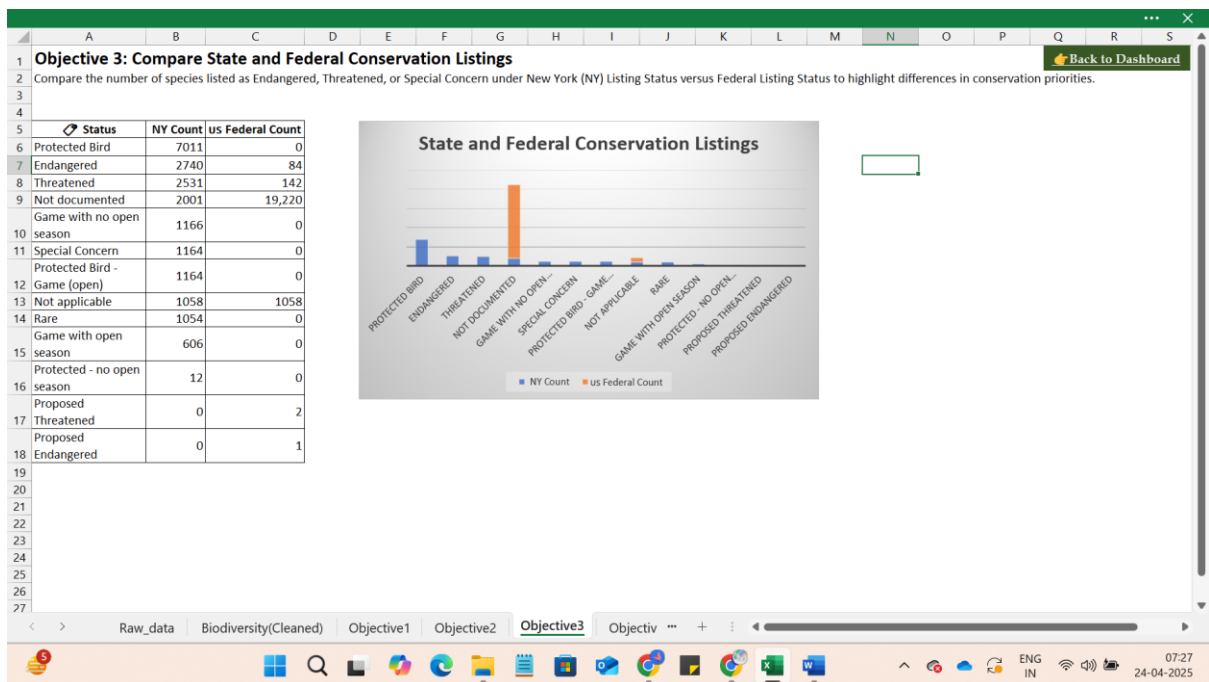
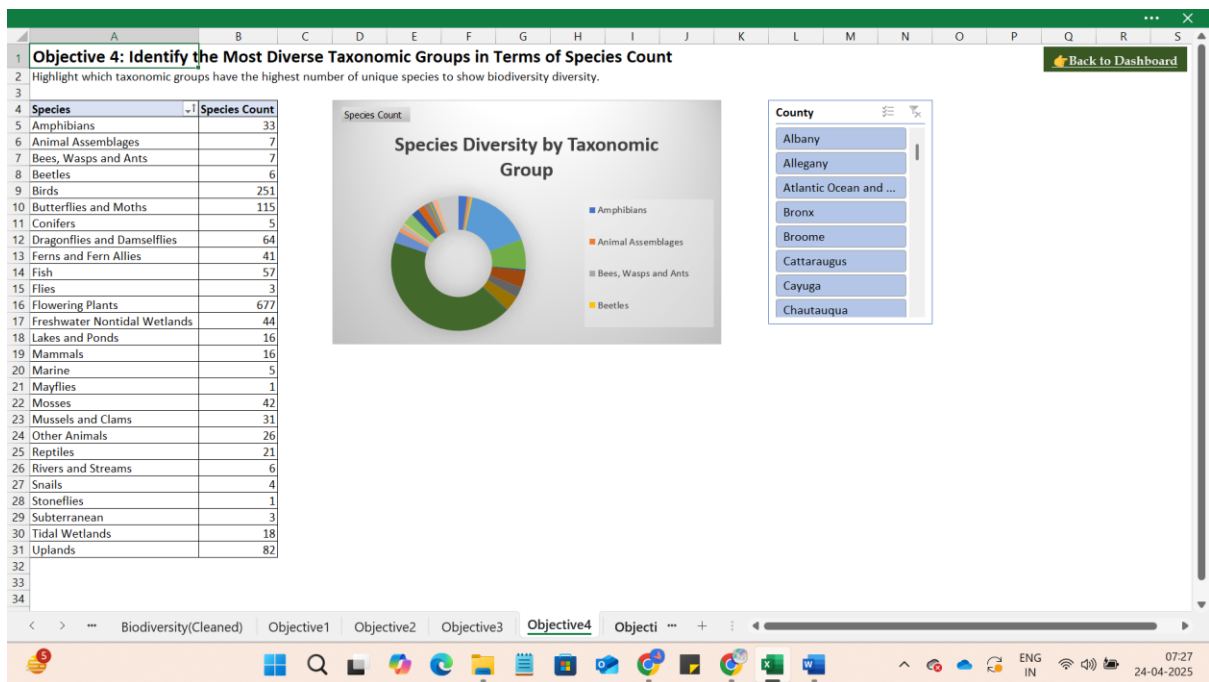
- High count of “Protected” and “Game with no open season” species at the state level.
- A distinct peak under the Game with no open season listing federally suggests targeted conservation efforts.

Global Conservation Status

- Most species fall under **G5 (Secure)**, followed by **G4 (Apparently Secure)** and **G3 (Vulnerable)**.
- Very few species are globally rare (**G1, G2**), but they need urgent attention.

Overall Sheets preview added in the document





	A	B	C	D	E	F	G	H	I	J	K
1	County	Category	Taxonomic Group	Taxonomic Subgroup	Scientific Name	Common Name	Year Last Documented	First Listed	Year (Numbers)	Threatening Status	Recovery Status
2	Albany	Animal	Amphibians	Frogs and Toads	Anaxyrus americanus	American Toad	1990-1999	1999	1999	Game with open season	Not documented
3	Albany	Animal	Amphibians	Frogs and Toads	Anaxyrus fowleri	Fowler's Toad	1990-1999	1999	1999	Game with open season	Not documented
4	Albany	Animal	Amphibians	Frogs and Toads	Hyla versicolor	Gray Treefrog	1990-1999	1999	1999	Game with open season	Not documented
5	Albany	Animal	Amphibians	Frogs and Toads	Lithobates catesbeianus	Bullfrog	1990-1999	1999	1999	Game with open season	Not documented
6	Albany	Animal	Amphibians	Frogs and Toads	Lithobates clamitans	Green Frog	1990-1999	1999	1999	Game with open season	Not documented
7	Albany	Animal	Amphibians	Frogs and Toads	Lithobates palustris	Pickering Frog	1990-1999	1999	1999	Game with open season	Not documented
8	Albany	Animal	Amphibians	Frogs and Toads	Lithobates pipiens	Northern Leopard Frog	1990-1999	1999	1999	Game with open season	Not documented
9	Albany	Animal	Amphibians	Frogs and Toads	Lithobates sylvaticus	Wood Frog	1990-1999	1999	1999	Game with open season	Not documented
10	Albany	Animal	Amphibians	Frogs and Toads	Pseudacris crucifer	Spring Peeper	1990-1999	1999	1999	Game with open season	Not documented
11	Albany	Animal	Amphibians	Frogs and Toads	Scaphiopus holbrookii	Eastern Spadefoot	2006-2006	2006	2006	Social Concern	Not documented
12	Dutchess	Animal	Birds	Hérons, Bitterns, Egrets, Pelicans	Ixobrychus exilis	Least Bittern	1990-1999	1999	2019-2019	Threatened	Not documented
13	Albany	Animal	Amphibians	Salamanders	Ambystoma jeffersonianum x laterale	Jefferson Salamander Complex	1990-1999	1999	1999	Game with no open seas	Not documented
14	Albany	Animal	Amphibians	Salamanders	Ambystoma laterale	Blue-spotted Salamander	1990-1999	1999	1999	Social Concern	Not documented
15	Albany	Animal	Amphibians	Salamanders	Ambystoma maculatum	Spotted Salamander	1990-1999	1999	1999	Game with no open seas	Not documented
16	Albany	Animal	Amphibians	Salamanders	Ambystoma tigrinum	Tiger Salamander	Unknown	Unknown	Unknown	Endangered	Not documented
17	Albany	Animal	Amphibians	Salamanders	Desmognathus fuscus	Dusky Salamander	1990-1999	1999	1999	Game with no open seas	Not documented
18	Albany	Animal	Amphibians	Salamanders	Desmognathus ochropneus	Allegheny Mountain Dusky Salamander	1990-1999	1999	1999	Game with no open seas	Not documented
19	Albany	Animal	Amphibians	Salamanders	Eurycea bislineata	Northern Two-lined Salamander	1990-1999	1999	1999	Game with no open seas	Not documented
20	Albany	Animal	Amphibians	Salamanders	Gyrinophilus porphyriticus	Spring Salamander	1990-1999	1999	1999	Game with no open seas	Not documented
21	Albany	Animal	Amphibians	Salamanders	Hemidactylum scutatum	Four-toed Salamander	1990-1999	1999	1999	Game with no open seas	Not documented
22	Albany	Animal	Amphibians	Salamanders	Notophthalmus viridescens	Eastern Newt	1990-1999	1999	1999	Game with no open seas	Not documented
23	Fulton	Animal	Birds	Nuthatches	Sitta carolinensis	White-breasted Nuthatch	2000-2005	2005	2005	Protected Bird	Not documented
24	Albany	Animal	Amphibians	Salamanders	Plethodon cinereus	Redback Salamander	1990-1999	1999	1999	Game with no open seas	Not documented
25	Albany	Animal	Amphibians	Salamanders	Plethodon glutinosus	Northern Slimy Salamander	1990-1999	1999	1999	Game with no open seas	Not documented
26	Albany	Animal	Amphibians	Salamanders	Pseudotriton ruber	Red Salamander	1990-1999	1999	1999	Game with no open seas	Not documented
27	Albany	Animal	Animal Assemblages	Animal Assemblages	Anadromous Fish Concentration Area	Anadromous Fish Concentration Area	1986-1986	1986	1986	Not applicable	Not applicable
28	Albany	Animal	Animal Assemblages	Animal Assemblages	Bat Colony	Bat Colony	2018-2018	2018	2018	Not applicable	Not applicable
29	Albany	Animal	Bees, Wasps and Ants	Bees	Bombus affinis	Rusty-patched Bumble Bee	1991-1991	1991	1991	Not documented	Endangered
30	Albany	Animal	Bees, Wasps and Ants	Bees	Bombus pennsylvanicus	American Bumble Bee	1987-1987	1987	1987	Not documented	Not documented
31	Albany	Animal	Bees, Wasps and Ants	Bees	Bombus terricola	Yellow-banded Bumble Bee	2009-2009	2009	2009	Not documented	Not documented
32	Albany	Animal	Birds	Blackbirds and Orioles	Agelaius phoeniceus	Red-winged Blackbird	2000-2005	2005	2005	Protected Bird	Not documented
33	Albany	Animal	Birds	Blackbirds and Orioles	Dolichonyx oryzivorus	Bobolink	2000-2005	2005	2005	Protected Bird	Not documented
34	Albany	Animal	Birds	Blackbirds and Orioles	Icterus galbula	Baltimore Oriole	2000-2005	2005	2005	Protected Bird	Not documented
35	Albany	Animal	Birds	Blackbirds and Orioles	Icterus spurius	Orchard Oriole	2000-2005	2005	2005	Protected Bird	Not documented
36	Albany	Animal	Birds	Blackbirds and Orioles	Molothrus ater	Brown-headed Cowbird	2000-2005	2005	2005	Protected Bird	Not documented
37	Albany	Animal	Birds	Blackbirds and Orioles	Quiscalus quiscula	Common Grackle	2000-2005	2005	2005	Protected Bird	Not documented
38	Albany	Animal	Birds	Blackbirds and Orioles	Sturnella magna	Eastern Meadowlark	2000-2005	2005	2005	Protected Bird	Not documented
39	Albany	Animal	Birds	Cardinals and Buntings	Cardinalis cardinalis	Northern Cardinal	2000-2005	2005	2005	Protected Bird	Not documented
40	Albany	Animal	Birds	Cardinals and Buntings	Passerina cyanea	Indigo Bunting	2000-2005	2005	2005	Protected Bird	Not documented
41	Albany	Animal	Birds	Cardinals and Buntings	Phoenicurus ludovicianus	Rose-breasted Grosbeak	2000-2005	2005	2005	Protected Bird	Not documented
42	Albany	Animal	Birds	Chickadees and Titmice	Piranga olivacea	Scarlet Tanager	2000-2005	2005	2005	Protected Bird	Not documented
43	Albany	Animal	Birds	Chickadees and Titmice	Baeolophus bicolor	Tufted Titmouse	2000-2005	2005	2005	Protected Bird	Not documented
44	Albany	Animal	Birds	Chickadees and Titmice	Parus atricapillus	Black-capped Chickadee	2000-2005	2005	2005	Protected Bird	Not documented
45	Albany	Animal	Birds	Cormorants	Phalacrocorax auritus	Double-crested Cormorant	2000-2005	2005	2005	Protected Bird	Not documented
46	Albany	Animal	Birds	Crows and Jays	Certhia americana	Brown Creeper	2000-2005	2005	2005	Protected Bird	Not documented
47	Albany	Animal	Birds	Crows and Jays	Corvus brachyrhynchos	American Crow	2000-2005	2005	2005	Protected Bird - Game w	Not documented
48	Albany	Animal	Birds	Crows and Jays	Corvus corax	Common Raven	2000-2005	2005	2005	Protected Bird	Not documented
49	Albany	Animal	Birds	Crows and Jays	Corvus ossifragus	Fish Crow	2000-2005	2005	2005	Protected Bird - Game w	Not documented
50	Albany	Animal	Birds	Crows and Jays	Corvus cristatus	Blue Jay	2000-2005	2005	2005	Protected Bird	Not documented
51	Albany	Animal	Birds	Cuckoos	Coccyus americanus	Yellow-billed Cuckoo	2000-2005	2005	2005	Protected Bird	Not documented
52	Albany	Animal	Birds	Cuckoos	Coccyus erythrophthalmus	Black-billed Cuckoo	2000-2005	2005	2005	Protected Bird	Not documented
53	Albany	Animal	Birds	Ducks, Geese, Waterfowl	Ala sponsa	Wood Duck	2000-2005	2005	2005	Protected Bird - Game w	Not documented
54	Albany	Animal	Birds	Ducks, Geese, Waterfowl	Blue-winged Teal	Blue-winged Teal	2000-2005	2005	2005	Protected Bird - Game w	Not documented

	A	B	C	D	E	F
1	County	Category	Taxonomic Group	Taxonomic Subgroup	Scientific Name	Common Name
2	Albany	Animal	Amphibians	Frogs and Toads	Anaxyrus americanus	American Toad
3	Albany	Animal	Amphibians	Frogs and Toads	Anaxyrus fowleri	Fowler's Toad
4	Albany	Animal	Amphibians	Frogs and Toads	Hyla versicolor	Gray Treefrog
5	Albany	Animal	Amphibians	Frogs and Toads	Lithobates catesbeianus	Bullfrog
6	Albany	Animal	Amphibians	Frogs and Toads	Lithobates clamitans	Green Frog
7	Albany	Animal	Amphibians	Frogs and Toads	Lithobates palustris	Pickering Frog
8	Albany	Animal	Amphibians	Frogs and Toads	Lithobates pipiens	Northern Leopard Frog
9	Albany	Animal	Amphibians	Frogs and Toads	Lithobates sylvaticus	Wood Frog
10	Albany	Animal	Amphibians	Frogs and Toads	Pseudacris crucifer	Spring Peeper
11	Albany	Animal	Amphibians	Frogs and Toads	Scaphiopus holbrookii	Eastern Spadefoot
12	Dutchess	Animal	Birds	Hérons, Bitterns, Egrets, Pelicans	Ixobrychus exilis	Least Bittern
13	Albany	Animal	Amphibians	Salamanders	Ambystoma jeffersonianum x laterale	Jefferson Salamander Complex
14	Albany	Animal	Amphibians	Salamanders	Ambystoma laterale	Blue-spotted Salamander
15	Albany	Animal	Amphibians	Salamanders	Ambystoma maculatum	Spotted Salamander
16	Albany	Animal	Amphibians	Salamanders	Ambystoma tigrinum	Tiger Salamander
17	Albany	Animal	Amphibians	Salamanders	Desmognathus fuscus	Dusky Salamander
18	Albany	Animal	Amphibians	Salamanders	Desmognathus ochropneus	Allegheny Mountain Dusky Salamander
19	Albany	Animal	Amphibians	Salamanders	Eurycea bislineata	Northern Two-lined Salamander
20	Albany	Animal	Amphibians	Salamanders	Gyrinophilus porphyriticus	Spring Salamander
21	Albany	Animal	Amphibians	Salamanders	Hemidactylum scutatum	Four-toed Salamander
22	Albany	Animal	Amphibians	Salamanders	Notophthalmus viridescens	Eastern Newt
23	Fulton	Animal	Birds	Nuthatches	Sitta carolinensis	White-breasted Nuthatch
24	Albany	Animal	Amphibians	Salamanders	Plethodon cinereus	Redback Salamander
25	Albany	Animal	Amphibians	Salamanders	Plethodon glutinosus	Northern Slimy Salamander
26	Albany	Animal	Amphibians	Salamanders	Pseudotriton ruber	Red Salamander
27	Albany	Animal	Animal Assemblages	Animal Assemblages	Anadromous Fish Concentration Area	Anadromous Fish Concentration Area
28	Albany	Animal	Animal Assemblages	Animal Assemblages	Bat Colony	Bat Colony
29	Albany	Animal	Bees, Wasps and Ants	Bees	Bombus affinis	Rusty-patched Bumble Bee
30	Albany	Animal	Bees, Wasps and Ants	Bees	Bombus pennsylvanicus	American Bumble Bee
31	Albany	Animal	Bees, Wasps and Ants	Bees	Bombus terricola	Yellow-banded Bumble Bee



5. CONCLUSION

This report has presented a comprehensive analysis of species diversity, conservation status, and documentation trends across New York State, offering valuable insights for ecological planning, habitat protection, and policy formulation. The findings underscore the importance of sustained biodiversity monitoring and the role of data visualization in identifying patterns critical to conservation efforts.

5.1 Key Findings and Observations

• Species Documentation Trends:

- Increased Recent Activity:** A significant surge in species documentation has occurred since the 1990s, with notable peaks in the early 2000s and again around 2019. This trend suggests a growing effort in recent decades to catalog and protect biodiversity, potentially due to technological advancements and heightened ecological awareness.
- Historical Gaps:** Prior to the 1980s, species documentation remained relatively low and stable, which could reflect either lower sampling efforts or limited awareness of biodiversity concerns during earlier decades.

• Conservation Status by County:

- High Concentration in Specific Counties:** Suffolk County recorded the highest number of species under conservation status, especially in categories such as "Protected - no open season" and "Threatened." Counties like Erie, Jefferson, and Orange also showed a consistently high count across multiple categories.
- Variety of Status Labels:** The New York State dataset distinguishes between various protection levels, including birds protected with or without open season, rare and

threatened species, and species of special concern—reflecting a nuanced approach to species management.

- **Taxonomic Group Diversity:**

- **Dominance of Butterflies, Moths, and Amphibians:** Species counts are highest among Butterflies and Moths, indicating their prevalence and vulnerability. Amphibians, birds, and beetles also show significant diversity, necessitating group-specific strategies.
- **Underrepresented Groups:** Certain groups like Animal Assemblages and Reptiles make up a smaller proportion of the total diversity, but their ecological significance still warrants focused attention.

- **Overall Distribution Patterns:**

- **Birds and Flowering Plants Lead:** A treemap view highlights that birds and flowering plants dominate the species landscape in New York, comprising the majority of documented organisms. This information can guide conservation efforts toward these dominant ecological categories while ensuring smaller groups are not overlooked.

- **Global Conservation Rankings (G1–G5):**

- **County-Level Variation:** Conservation status grades (G1 to G5) show wide distribution across counties, with counties like Seneca and Sullivan having a relatively higher presence of species ranked as G1 (critically imperiled) or G2 (imperiled).
- **Prevalence of G5:** The majority of species are listed as G5 (secure), but the presence of G1–G3 species across all counties emphasizes the need for localized conservation planning.

5.2 Role of Slicers and Filters in the Biodiversity Dashboard

The Biodiversity Dashboard built in Microsoft Excel utilizes **interactive slicers** and filters to empower users with dynamic data exploration capabilities.

- **Slicers:**

Slicers enable users to filter the data by:

- **County:** Drill down into species data on a localized level.
- **NY Listing Status:** Quickly assess the protection category and urgency for various species.
- **Taxonomic Group:** Compare representation and distribution of amphibians, birds, insects, plants, and more.
- **Conservation Category (G1–G5):** Focus on species based on global threat levels.

These slicers provide a user-friendly interface for stakeholders to analyze data across multiple dimensions with ease.

5.3 Combined Observations and Implications

- **Localized Biodiversity Hotspots:** Counties like Suffolk and Erie serve as biodiversity hotspots, hosting high numbers of both protected and imperiled species, thereby warranting targeted monitoring and conservation efforts.
- **Need for Multi-Tiered Conservation Strategy:** The mix of state and federal protection categories along with global conservation grades (G1–G5) calls for coordination across governmental levels and NGOs to address species needs effectively.
- **Data-Driven Ecology:** The use of dashboards and historical documentation trends provides strong foundations for evidence-based environmental policies, funding decisions, and habitat restoration initiatives.

5.4 Recommendations

- **Enhance Monitoring in High-Biodiversity Counties:** Prioritize conservation and documentation efforts in counties with a high number of threatened or rare species.
- **Invest in Understudied Taxonomic Groups:** Allocate research resources to lesser-documented groups like Animal Assemblages and Reptiles to close biodiversity knowledge gaps.
- **Develop Localized Conservation Action Plans:** Use G1–G3 data to inform county-specific action plans that address critically imperiled species.
- **Promote Citizen Science and Public Engagement:** Engage local communities in biodiversity documentation and habitat stewardship to supplement formal conservation efforts.
- **Integrate State and Federal Conservation Strategies:** Align listing statuses across state and federal levels for cohesive and enforceable conservation policy.
- **Update and Maintain Dynamic Dashboards:** Continue refining interactive dashboards to support real-time decision-making and adaptive management strategies.

6. FUTURE SCOPE

The present analysis of biodiversity documentation and conservation status across New York State establishes a strong foundation for future research, technological development, and strategic policy-making in environmental conservation. Several promising directions exist to extend the impact of this project, enhance ecological understanding, and support decision-makers and stakeholders. Future work can focus on the following areas:

6.1 Enhanced Data Integration

- **Expand Environmental Datasets:**

Integrate additional datasets to build a more comprehensive understanding of biodiversity and conservation challenges:

- **Habitat and Land Use Data:** Incorporate spatial data on land cover, protected areas, deforestation, and urbanization to understand habitat fragmentation and its impacts.
- **Climate and Pollution Data:** Use temperature, precipitation, and air/water quality data to examine environmental stressors influencing species distribution and survival.
- **Citizen Science Contributions:** Integrate data from platforms like iNaturalist or eBird to capture real-time observations and enrich documentation efforts.
- **Invasive Species and Disease Records:** Link to databases tracking invasive species or wildlife diseases to assess emerging threats to native biodiversity.

- **Real-Time Monitoring Capabilities:**

Explore tools like remote sensing and sensor networks (e.g., for water quality or animal movement) to enable real-time tracking of environmental indicators and species health.

6.2 Advanced Analytical Techniques

- **Predictive Ecological Modeling:** Develop models to forecast species vulnerability, habitat degradation, or extinction risk under scenarios such as climate change, urban expansion, or policy shifts.

- **Spatial Analysis and Biodiversity Mapping:** Use GIS tools and spatial statistics to identify conservation hotspots, migration corridors, and under-surveyed regions. Methods may include:

- Species distribution modeling (SDM)
- Kernel density estimation for rare species
- Gap analysis for conservation planning

- **Temporal Trend Analysis:** Apply time-series methods to analyze changes in biodiversity documentation over decades, helping identify the impact of policy actions, ecological events, or public awareness efforts.

- **Taxonomic Imbalance Assessment:** Use statistical tests to identify potential sampling biases across taxonomic groups, helping guide future documentation priorities.

6.3 Development of an Interactive Web Application

- **Public-Facing Biodiversity Dashboard:** Create a web-based platform to democratize access to biodiversity data and visualizations, tailored for researchers, policymakers, educators, and the public. Key features could include:

- Interactive maps showing conservation status by county and taxonomic group
- Species search tools with image galleries and habitat information
- Filters by conservation ranking (G1–G5), NY listing status, and time periods

- **Custom Visual Analytics:** Enable users to generate downloadable reports, select time periods of interest, and view longitudinal or comparative species data at the county level.

- **Mobile and Field Accessibility:** Ensure mobile-friendly access so that field researchers and conservation volunteers can use the tool in real-time and in remote locations.

6.4 Community Engagement and Environmental Education

- **Biodiversity Literacy Campaigns:** Develop educational resources—including infographics, biodiversity fact sheets, and regional species guides—to raise public awareness on local wildlife and conservation needs.

- **Citizen Reporting Tools:** Allow the public to submit sightings or photos of species directly into the dashboard to enrich the dataset and foster community involvement.
- **Partnerships with Schools and Universities:** Encourage the use of biodiversity data in school curricula and student-led ecological monitoring programs to build a new generation of environmental stewards.
- **Feedback Loops and Surveys:** Create channels for environmental groups and local communities to provide feedback on the usability and focus of the dashboard and suggest new research priorities.

6.5 Policy Implications and Evaluation

- **Data-Driven Conservation Planning:** Use findings to recommend zoning changes, protected area designations, or habitat restoration in counties with high concentrations of vulnerable species (G1–G3).
- **Targeted Funding Allocation:** Advocate for increased funding in counties and species groups that are under-documented but ecologically significant.
- **Policy Impact Tracking:** Monitor the effectiveness of conservation policies over time using updated biodiversity documentation data and evaluate whether target species show recovery or decline.
- **Collaborative Conservation Frameworks:** Facilitate cross-county or regional conservation initiatives by sharing data, trends, and maps with state and federal agencies, non-profits, and academic partners.

By pursuing these forward-looking strategies, this project can evolve into a dynamic and impactful tool for safeguarding biodiversity in New York. It will not only strengthen the scientific understanding of local ecosystems but also empower communities and institutions to take meaningful action toward conservation and sustainability.

7. REFERENCES

To ensure the accuracy, reliability, and relevance of this report on species documentation and conservation status in New York, the following sources were consulted:

For dataset: <https://catalog.data.gov/dataset/biodiversity-by-county-distribution-of-animals-plants-and-natural-communities>

Microsoft Excel Support – PivotTables, Data Visualization, and Dashboards

- Overview of PivotTables and Pivot Charts: <https://support.microsoft.com/en-us/office/527c8fa3-02c0-445a-a2db-7794676bce96>
- Getting Started with Dashboards: <https://support.microsoft.com/en-us/office/edc6a32c-85c5-443d-9103-56dcc7bb24a4>
- Top Ten Ways to Clean Your Data: <https://support.microsoft.com/en-us/office/2844b620-677c-47a7-ac3e-c2e157d1db19>
- Analyze Data in Excel: <https://support.microsoft.com/en-us/office/3223aab8-f543-4fda-85ed-76bb0295ffc4>

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